EDUVISIONAI : AI-POWERED HISTORY E-LEARNING SYSTEM FOR VISUALLY IMPAIRED O-LEVEL STUDENTS IN SRI LANKA

25-26J-438

Project Proposal Report

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B.Sc. (Hons) in Information Technology Specializing in

Software Engineering

Department of Computer Science & Software Engineering

Sri Lanka Institute of Information Technology

Sri Lanka

August 2025

EDUVISIONAI : AI-POWERED HISTORY E-LEARNING SYSTEM FOR VISUALLY IMPAIRED O-LEVEL STUDENTS IN SRI LANKA

AI-Powered Braille-to-Text Conversion and Automated Evaluation System for O/L History Examinations

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Declaration

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously publish or written by another person expect where the acknowledgement is made in the text.

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ABSTRACT

Preparation for Ordinary Level (O/L) History examinations requires students to practice structured and essay-type questions, which demand factual knowledge, critical reasoning, and contextual understanding. For visually impaired learners who rely on Braille, this process presents significant challenges. Traditional correction methods require teachers to manually evaluate Braille responses, a task that is time-consuming and resource-intensive. This reliance on manual evaluation limits opportunities for frequent practice, reduces timely feedback, and hampers students' ability to engage in self-directed learning, placing visually impaired students at a disadvantage compared to their sighted peers.

To address these challenges, this project introduces **EduVisionAI**, an automated correction platform designed for O/L History exam preparation in Braille. The system workflow begins when a student uploads a Braille exam sheet, which is converted into a digital, machine-readable format using Braille-to-text conversion techniques. These digitized responses are then evaluated against a trained answer model built from past examination papers, government-approved textbooks, and official marking schemes. By applying natural language processing (NLP) and semantic similarity methods, EduVisionAI assesses the accuracy, relevance, structure, and completeness of student answers.

Beyond scoring, EduVisionAI generates detailed, personalized feedback, highlighting factual errors, missing historical context, and suggestions for improved argumentation and essay structure. Results are provided in both text and audio formats, ensuring accessibility and enabling independent review.

By combining Braille recognition, automated grading, and multimodal feedback delivery, EduVisionAI addresses a critical gap in inclusive education. It reduces reliance on manual correction, provides immediate and consistent evaluation, and empowers visually impaired students to practice independently. Ultimately, the system enhances exam readiness, fosters self-assessment, and promotes equitable access to education within the Sri Lankan O/L context. The framework also holds potential for expansion to other subjects and examination levels, offering a scalable solution for accessible learning.

Keywords: Braille-to-Text Conversion, Ordinary Level (O/L) History, Automated Exam Correction, Natural Language Processing (NLP), Automated Grading System, Accessible Education, Visually Impaired Learners, Inclusive Learning Technologies

List of Abbreviations

OCR	Optical Character Recognition
NLP	Natural Language Processing
ML	Machine Learning
AI	Artificial Intelligence
IDE	Integrated Development Environment
VS Code	Visual Studio Code
O/L	Ordinary Level
TTS	Text-to-Speech
DB	Database
API	Application Programming Interface

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1 Introduction

1.1 Background and Literature Review

History is a key subject in the Ordinary Level (O/L) curriculum, requiring students to answer structured and essay-type questions that assess not only factual knowledge but also the ability to critically analyze events, provide historical context, and construct well-organized arguments. For visually impaired learners, preparing for O/L History examinations poses significant challenges, as they rely on Braille to read and write their answers. Unlike sighted students, who can easily access textbooks, model papers, and online resources, visually impaired learners face barriers in both accessing study materials and receiving feedback on their written responses.

Traditional correction methods for Braille exam scripts depend heavily on manual evaluation by teachers or assistants. This process is time-consuming, resource-intensive, and often inconsistent, making it difficult for students to practice independently or receive timely feedback. Without frequent feedback, students struggle to identify weaknesses in their answers, limiting their ability to self-assess and improve. This lack of independent practice opportunities creates a disadvantage in exam readiness and contributes to an educational gap between visually impaired and sighted learners.

In recent years, research in Braille recognition **systems** has shown promising results in converting Braille text into machine-readable formats, enabling integration with digital platforms for accessibility [1][2]. Similarly, advances in Natural Language Processing (NLP) and automated essay scoring (AES) have demonstrated the potential of AI-powered systems to evaluate descriptive answers, assess semantic similarity with model answers, and generate feedback for student learning [3][4]. However, most existing tools are designed for general-purpose education or essay scoring and are not tailored to the specific requirements of the Sri Lankan O/L History marking scheme.

This research builds upon these developments by proposing EduVisionAI, an AI-powered platform that integrates Braille-to-text conversion with automated answer evaluation specifically for O/L History. By leveraging NLP techniques, semantic similarity algorithms, and government-

approved marking schemes, EduVisionAI is designed to provide accurate scoring, highlight factual or contextual errors, and deliver constructive feedback in both text and audio formats.

Advantages of the Proposed System

The proposed system offers several advantages over traditional manual correction methods:

- Accessibility for Visually Impaired Students – Converts Braille exam sheets into digital text, making automated evaluation possible without manual intervention.
- Automated and Consistent Evaluation – Uses trained answer models aligned with the O/L History syllabus and marking scheme, ensuring fairness and reducing human error.
- Personalized Feedback Generation – Highlights factual mistakes, missing historical context, and weak argumentation, while suggesting ways to improve.
- Multimodal Output Delivery – Provides results in both text and audio formats, enabling visually impaired learners to engage with feedback independently.
- Scalability and Inclusivity – While initially developed for O/L History, the system can be extended to other subjects and examination levels, contributing to a broader framework for inclusive education.

1.2 **Research Gap**

The traditional method of correcting Braille examination papers is fully manual, requiring teachers or assistants to read and interpret Braille responses before applying the marking scheme. This process is time-consuming, labor-intensive, and inconsistent, as it depends heavily on human interpretation. More importantly, it limits opportunities for frequent practice and timely feedback, which are essential for developing exam readiness. As a result, visually impaired students are at a disadvantage compared to their sighted peers, who can access printed or digital automated marking resources.

Existing Braille recognition systems mainly focus on Braille-to-text translation, helping convert Braille into machine-readable formats. While these systems improve accessibility, they are not designed to evaluate or grade student answers, and they lack the ability to align with the Ordinary Level (O/L) History marking scheme used in Sri Lanka. On the other hand, research in automated

essay scoring (AES) and natural language processing (NLP) has shown success in grading descriptive answers in general education. However, most AES systems are domain-agnostic and do not incorporate subject-specific requirements such as historical accuracy, contextual relevance, and structured argumentation, which are essential for History exams.

Importantly, there is currently no automated correction system that supports the Sri Lankan O/L syllabus in Braille. To overcome this limitation, EduVisionAI integrates Braille recognition, NLP-driven evaluation, and subject-specific marking schemes into a unified system. This novel approach ensures that History answers written in Braille can be digitized, semantically analyzed, scored, and fed back to the student in both text and audio formats. By doing so, the proposed system fills the gap between general Braille accessibility tools and automated grading platforms, providing an inclusive solution tailored for O/L History in Sri Lanka.

Table 1 Research gap

Method	Strengths	Research Gaps
Manual Braille Correction	Accurate when done by experienced teachers; allows subjective judgment	Time-consuming and resource-intensive Limited opportunities for frequent practice Inconsistent marking depending on evaluator
Braille-to-Text Conversion Tools	Enables digital access to Braille documents; useful for reading support	No automated evaluation Cannot align answers with marking schemes Only translation, no feedback generation Not designed for the Sri Lankan O/L syllabus
Generic Automated Essay Scoring (AES)	NLP models can grade essays and provide basic scoring; scalable	Domain-agnostic, not tailored to History Fails to assess historical accuracy, contextual relevance, or structured argumentation No adaptation to Sri Lankan O/L curriculum

General NLP-based Education Tools	Strong text analysis capabilities; adaptive feedback possible	Designed for generic essays or comprehension tasks Lack of integration with Braille input No multimodal accessibility (e.g., audio feedback) Not syllabus-specific

1.3 Research Problem

In the Sri Lankan Ordinary Level (O/L) examination system, History is a compulsory subject that requires students to answer structured and essay-type questions. For visually impaired learners, preparing for these examinations is particularly challenging as they rely on Braille to write their answers. The evaluation of Braille exam scripts is traditionally manual, requiring teachers or assistants to interpret Braille responses and apply the marking scheme. This process is time-consuming, resource-intensive, and inconsistent, often limiting opportunities for regular practice and timely feedback. As a result, visually impaired students face disadvantages in developing self-assessment skills and achieving exam readiness compared to their sighted peers.

Existing Braille recognition systems focus mainly on Braille-to-text translation, which improves accessibility but does not extend to automated evaluation or alignment with the Sri Lankan O/L History syllabus. On the other hand, research in automated essay scoring (AES) and natural language processing (NLP) has demonstrated potential for grading descriptive answers in general education contexts. However, most of these solutions are domain-agnostic, designed for English essays or comprehension tasks, and do not incorporate subject-specific requirements such as historical accuracy, contextual relevance, or structured argumentation. Furthermore, none of the existing systems provide feedback tailored to the O/L History marking scheme used in Sri Lanka, nor do they offer multimodal accessibility features such as audio-based feedback delivery. The absence of an integrated solution that combines Braille recognition, automated grading, and syllabus-specific feedback creates a significant gap in supporting visually impaired O/L History students. Without such a system, students remain heavily dependent on teachers for corrections and lack the opportunity to practice independently, receive consistent evaluation, and build confidence in their exam preparation.

2 Objectives

2.1 Main Project

To design and develop EduVisionAI, an automated Braille-based exam paper correction system tailored for the Sri Lankan O/L History syllabus, which digitizes Braille exam responses, evaluates them using NLP-driven trained models and official marking schemes, and generates personalized feedback in both text and audio formats, thereby empowering visually impaired students to practice independently and improve exam readiness.

2.2 Specific Objectives

- Implement Braille-to-Text Conversion – Develop a Braille recognition module to digitize uploaded Braille exam sheets into machine-readable text.
- Develop Trained Answer Model – Build a model trained on past O/L History exam papers, government-approved textbooks, and marking schemes to serve as the evaluation benchmark.
- Automated Answer Evaluation – Apply NLP techniques and semantic similarity analysis to compare student answers with the trained model answers for accuracy, relevance, and completeness.
- Feedback Generation – Provide detailed, constructive feedback highlighting factual errors, missing historical context, weak argumentation, and improvement suggestions.
- Multimodal Feedback Delivery – Deliver results in both text and audio (Text-to-Speech) formats to ensure accessibility for visually impaired learners.
- System Integration & Testing – Integrate all modules into a unified platform and evaluate system performance in terms of accuracy, efficiency, accessibility, and usability.
- Scalability Consideration – Ensure the system can be extended to other subjects or examination levels in the future, promoting long-term educational inclusivity.

3 Methodology

3.1 Requirement Gathering

The proposed methodology focuses on developing EduVisionAI, an automated Braille exam correction system specifically designed for the Sri Lankan O/L History subject. The system integrates Braille-to-text recognition, NLP-based automated grading, and multimodal feedback delivery.

3.1.1 Dataset collection

To ensure robust evaluation and accurate feedback, a multi-source dataset acquisition strategy will be followed:

- Past O/L History Examination Papers – Collect past structured and essay-type questions along with their marking schemes issued by the Department of Examinations, Sri Lanka.
- Government-Approved History Textbooks – Include prescribed textbooks and resource materials for fact-checking and training answer models.
- Model Answers and Teacher Marking Guides – Gather answer scripts evaluated by experienced History teachers to fine-tune the answer evaluation module.
- Braille Input Samples – Collect Braille-written responses from visually impaired students to train and test the Braille-to-text conversion module.

3.2 Development Methodology

The development is divided into several modules, each focusing on a core functionality of EduVisionAI .

3.2.1 Braille-to-Text Conversion Module

Description:
The Braille-to-Text Conversion Module is the entry point of EduVisionAI, designed to handle uploaded scanned PDF exam sheets written in Braille. Since visually impaired students typically submit answers in Braille script, this module converts the scanned Braille responses into digital, machine-readable text suitable for automated evaluation.

The workflow of this module includes:

1. PDF Upload & Extraction – Each page of the PDF is converted into images for processing.
2. Preprocessing – Grayscale conversion, noise reduction, thresholding, skew correction, and segmentation are applied to enhance recognition.
3. Braille OCR Recognition – A Braille-trained Tesseract OCR or custom deep learning model maps Braille cells into characters, with special handling for punctuation and formatting.
4. Post-Processing – Recognized text is corrected, structured, and prepared for evaluation.

Technologies & Tools:

- Tesseract OCR (Braille support)
- Python (PyPDF2, pdf2image, OpenCV, NumPy)
- Google Colab / Jupyter Notebook for testing

3.2.2 Trained Answer Model Module

Description:
This module creates a reference answer model for evaluating O/L History exam responses. A curated dataset is built using past examination papers, government-approved textbooks, and official marking schemes. The model stores structured answers, key historical facts, and essay patterns (introductions, argument flow, and conclusions) that serve as the benchmark for automated evaluation. When a student's converted text is submitted, it is compared against these trained reference answers using semantic similarity techniques to assess factual accuracy, contextual relevance, and completeness.

Technologies & Tools:

- Python – for data preprocessing, tokenization, and dataset preparation.
- MongoDB / SQL – to store model answers, marking schemes, and structured essay templates.
- Hugging Face Transformers – for text embeddings and semantic similarity analysis between student answers and reference models.

- Kaggle Datasets – as supplementary resources for training NLP models, such as essay scoring datasets and text similarity corpora, to enhance model robustness.

3.2.3 Automated Evaluation & NLP Module

Description:
This module is responsible for the core grading process of EduVisionAI. Once the student's Braille responses have been converted into text, they are compared against the trained answer model using advanced NLP techniques. The system applies semantic similarity analysis, keyword matching, and contextual understanding to measure how closely a student's response aligns with expected answers. In addition to simple fact-checking, the module evaluates historical accuracy (dates, events, figures), relevance to the question, logical flow, and completeness of arguments. The results include both a numerical score and qualitative feedback highlighting strengths and areas for improvement.

Technologies & Tools:

- NLP Models (BERT, RoBERTa, GPT-based fine-tuned models) – for contextual text understanding and semantic evaluation.
- Scikit-learn & TensorFlow – for model training, classification, and performance evaluation.
- Sentence Transformers – to perform semantic similarity analysis between student responses and model answers.

3.2.4 Feedback Generation Module

Description:
The Feedback Generation Module delivers personalized, constructive feedback to help students improve their History exam performance. Instead of only providing numerical marks, this module highlights factual inaccuracies, missing historical details, weak argumentation, and structural issues in student answers. It also suggests improvements in clarity, organization, and historical reasoning, ensuring alignment with the official O/L History marking scheme. Feedback is

generated in both text and audio formats, enabling visually impaired learners to access it independently.

Technologies & Tools:

- Python – for implementing rule-based scoring, error detection, and integration with evaluation results.
- NLP Summarization Techniques – to generate concise, human-like feedback tailored to each student's response.
- Knowledge Base of History Marking Criteria – a structured set of evaluation rules and guidelines to ensure feedback matches the Sri Lankan O/L examination standards.

3.2.5 Multimodal Feedback Delivery Module

Description:
This module is designed to make feedback accessible to all students, including those who are visually impaired. It delivers corrections and feedback in both text and audio formats. Guardians or teachers can read the feedback in text form, while students can listen to it via audio output. This dual-mode approach ensures that learning and evaluation remain inclusive, supporting different accessibility needs and learning preferences.

Technologies & Tools:

- Google TTS / pyttsx3: For converting textual feedback into audio, enabling students to listen to their corrections.
- React.js: To create an interactive and user-friendly interface where both text and audio feedback can be accessed seamlessly.
- Flask / FastAPI: For backend integration, handling requests for feedback generation, text-to-speech processing, and delivering the results to the frontend.

3.2.6 System Integration & Testing Module

Description:

This module focuses on integrating all individual components of the system into a single, cohesive platform. Once integrated, comprehensive system testing will be conducted to ensure the platform works reliably and efficiently. Key aspects of testing will include:

- Braille recognition accuracy: Ensuring that Braille inputs are correctly interpreted by the system.
- Answer evaluation reliability: Verifying that student responses are evaluated consistently and accurately.
- Feedback clarity: Checking that both text and audio feedback are clear, understandable, and actionable.
- Accessibility features: Confirming that features intended for visually impaired students, such as text-to-speech and navigation support, function as intended.

Technologies & Tools:

- Docker: To containerize the application, making deployment and environment setup consistent across platforms.
- Postman: For testing and validating APIs between modules, ensuring smooth communication and data exchange.
- Unit testing frameworks (PyTest / Jest): To perform automated testing of individual modules, catching errors early and maintaining system reliability.

3.3 Project Management Technology

Research requires effective project management, where scope, time, and cost must be carefully aligned through a structured system or procedure. For the adaptive quiz platform project, a process-oriented methodology will be employed, dividing the project into distinct stages with meticulously planned inputs and outputs. This approach is designed to accommodate changes, identify risks, and manage the project effectively. JIRA will be used to support this methodology, providing a robust platform for tracking progress, managing tasks, and ensuring that project requirements and timelines are met efficiently.

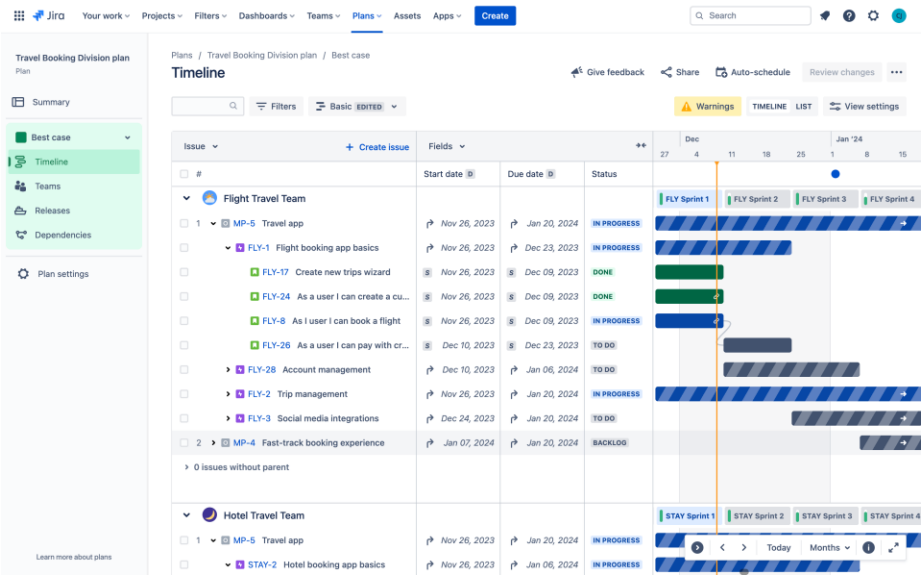

Figure 1Jira Dashboard

3.4 Feasibility Study

The feasibility study evaluates the practicality of implementing EduVisionAI, an automated correction platform for visually impaired students preparing for Ordinary Level (O/L) History examinations. This assessment considers technical capabilities, project scheduling, and economic viability, ensuring that the system can be developed and deployed successfully.

3.4.1 Technical Feasibility

Knowledge on Technologies

- Braille Recognition: Proficiency in Braille-to-text conversion techniques for accurate digitization of student responses.
- Natural Language Processing (NLP): Expertise in semantic similarity, text analysis, and automated assessment using libraries such as SpaCy, NLTK, and Transformer models.
- Automated Evaluation: Experience in scoring and evaluating structured and essay-type responses using trained models and historical exam data.
- Text-to-Speech (TTS): Knowledge of Google TTS or pyttsx3 for converting textual feedback into accessible audio for visually impaired learners.
- Web Application Development: Proficiency in React.js for frontend development and Flask/FastAPI for backend integration.

Knowledge on Tools

- Development Tools: VS Code, Jupyter Notebook, Google Colab for model experimentation and testing.
- Project Management: Jira for sprint planning, task tracking, and team collaboration.
- Dataset Annotation & Preprocessing Tools: Tools for labeling and preprocessing historical exam answers to build robust evaluation models.
- Dashboard & Visualization Tools: Flask/FastAPI, React.js, Chart.js/D3.js for creating interactive feedback dashboards.

Data Collection & Preprocessing Skills

- Collecting datasets from past O/L History papers, government-approved textbooks, and sample student responses.
- Preprocessing textual data for NLP tasks, including tokenization, normalization, and semantic analysis.
- Preparing audio output for accessibility, ensuring clarity and comprehensibility for visually impaired students.

3.4.2 Schedule Feasibility

The project will follow an Agile methodology, divided into 2–3 week sprints for each module:

- Sprint 1: Dataset collection and preprocessing
- Sprint 2: Braille-to-text conversion module development
- Sprint 3: Automated answer evaluation model training
- Sprint 4: Feedback generation module development
- Sprint 5: Text-to-speech integration for audio feedback
- Sprint 6: Frontend UI and dashboard development (React.js)
- Sprint 7: Backend integration (Flask/FastAPI)
- Sprint 8: System testing, optimization, and accessibility validation

Given the modular approach, the timeline is realistic and allows flexibility for refinement based on test feedback. Milestones will be tracked in Jira, ensuring tasks are delivered within deadlines.

3.5 System Analysis

3.5.1 Software Solution Approach

The proposed system, EduVisionAI, will be developed as a modular, multimodal feedback platform for visually impaired students preparing for O/L History examinations. The system integrates Braille recognition, automated answer evaluation, and multimodal feedback delivery (text and audio) to provide inclusive, accessible, and timely assessment.

Workflow:

1. Data Collection – Gather datasets including historical exam papers, official marking schemes, and sample student answers.
2. Pre-processing – Digitize Braille responses, normalize text, and prepare evaluation datasets.

3. Answer Evaluation Model – Use NLP techniques (semantic similarity, text analysis, and transformer models) to assess accuracy, relevance, structure, and completeness of student answers.

4. Feedback Generation – Generate detailed, personalized feedback highlighting factual errors, missing context, and suggestions for improving essay structure.

5. Multimodal Delivery – Convert textual feedback into audio using TTS (Google TTS / pyttsx3) to ensure accessibility for visually impaired learners.

6. Frontend Dashboard – Present results in an interactive dashboard allowing guardians to read feedback and students to listen to audio.

7. Integration and Testing – Combine all modules, test for reliability, and validate accessibility features to ensure system robustness.

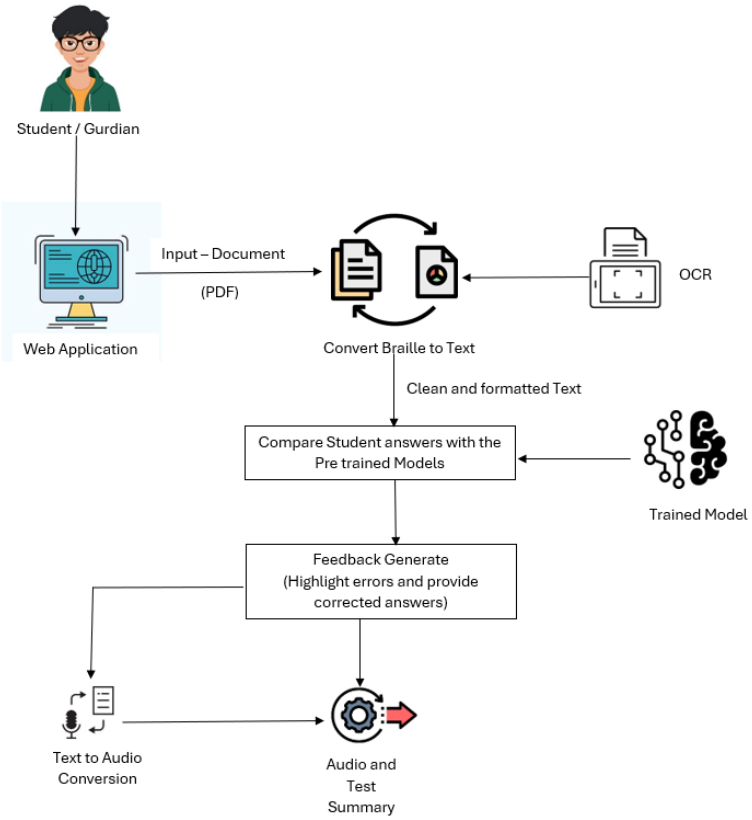

Figure 2 System Diagram

3.5.2 Tools and Technologies

The proposed system will use a combination of programming languages, frameworks, and tools to ensure robustness, scalability, and accessibility.

Programming Languages:

- Python 3.x – Core development language for NLP, Braille-to-text conversion, feedback generation, and TTS integration.
- JavaScript (React.js) – For building the responsive and interactive dashboard frontend.

Frameworks and Libraries:

- SpaCy / NLTK / Transformers – For text processing, semantic similarity, and automated answer evaluation.
- Pandas / NumPy – For data manipulation and preprocessing.
- Google TTS / pyttsx3 – For converting textual feedback into audio format.
- Flask / FastAPI – Backend web frameworks for RESTful API development and integration with frontend.

Data Collection and Preprocessing Tools:

- Past O/L History Papers & Official Textbooks – For building training datasets and evaluation models.
- Annotation Tools – Manual annotation of sample answers for supervised training and model validation.

Frontend Development and Visualization Tools:

- React.js – For creating a real-time, user-friendly dashboard to present feedback in both text and audio formats.
- Chart.js / D3.js – For visual representation of scores, progress, and common error trends.

Development and Project Management Tools:

- VS Code / Jupyter Notebook / Google Colab – For development, experimentation, and GPU-enabled NLP model training.
- Jira – For Agile sprint planning, backlog management, and task tracking.
- Git & GitHub – For version control and collaborative development.

Testing Tools:

- Pytest – For automated unit and integration testing of Python modules.
- Postman – For testing API endpoints between backend and frontend.

Hardware Requirements:

- GPU-enabled Workstation / Cloud GPU – For training NLP models efficiently.
- Braille Input Devices / Scanners – For capturing Braille exam sheets.
- Speakers / Headphones – For delivering audio feedback to students.

3.6 Project Requirements

The project requirements define the expected functionality and operational qualities of EduVisionAI, the automated O/L History correction and feedback system for visually impaired students. These requirements ensure that the system meets both performance goals and accessibility standards.

3.6.1 Functional Requirements

The system shall:

- Braille-to-Text Conversion – Accurately convert uploaded Braille exam sheets into machine-readable text.
- Automated Answer Evaluation – Evaluate structured and essay-type responses using NLP-based semantic analysis against a trained answer model.

- Feedback Generation – Generate detailed, personalized feedback highlighting factual errors, missing context, and suggestions for improving essay structure.
- Text-to-Speech (TTS) Delivery – Convert textual feedback into audio format using TTS for visually impaired students.
- Dashboard Visualization – Display results, scores, and feedback in a user-friendly dashboard for guardians and students.
- Progress Tracking – Maintain historical performance data to track student improvement over time.
- Module Integration – Ensure seamless interaction between Braille recognition, answer evaluation, feedback generation, and dashboard delivery.
- Data Logging – Store digitized responses, evaluation results, and feedback for reporting and model improvement.
- Accessibility Compliance – Support screen readers and other assistive technologies to ensure usability for visually impaired learners.

3.6.2 Non-Functional Requirements

The system shall:

- Performance – Process Braille uploads, evaluate answers, and generate feedback within 2–3 seconds per response.
- Accuracy – Provide evaluation results that closely match manual grading standards with high semantic similarity.
- Scalability – Support addition of other subjects and examination levels with minimal code modifications.
- Reliability – Maintain consistent performance across different types of Braille sheets, handwriting styles, and content complexity.

- Security – Ensure student data and feedback are stored securely and transmitted over encrypted channels.
- Maintainability – Have a modular code structure with comprehensive documentation for easy updates and debugging.
- Usability – Provide an intuitive interface accessible to students, teachers, and guardians without requiring technical expertise.
- Compatibility – Deployable on local systems or cloud-based environments with GPU-enabled or standard hardware.
- Resource Efficiency – Optimize computational resources to handle multiple concurrent evaluations without performance degradation.

3.7 Project Scope

The scope of this project defines the boundaries and extent of EduVisionAI, outlining the features and functionalities that will be developed in the current phase and those that are not included.

3.7.1 In-Scope

The system will include:

- Braille-to-Text Conversion – Convert uploaded Braille exam sheets into machine-readable digital text.
- Automated Answer Evaluation – Assess structured and essay-type responses using NLP-based semantic similarity techniques against a trained answer model.
- Feedback Generation – Provide detailed, personalized feedback highlighting factual errors, missing context, and suggestions for improving essay and answer structure.

- Text-to-Speech (TTS) Delivery – Convert textual feedback into audio for visually impaired students.
- Dashboard Visualization – Display results, scores, and feedback in a responsive web-based dashboard accessible to both students and guardians.
- Progress Tracking – Maintain a history of student performance for monitoring improvement over time.
- Module Integration – Ensure seamless interaction between Braille recognition, automated evaluation, feedback generation, and dashboard delivery.
- Data Logging – Store digitized answers, evaluation results, and feedback securely for reporting and model improvement.
- Accessibility Compliance – Support screen readers and other assistive technologies to ensure usability for visually impaired learners.
- Testing & Optimization – Evaluate system performance under varying student answer complexity and Braille input styles.

3.7.2 Out-of-Scope

The system will not include:

- Real-World Classroom Deployment – The system will be tested in controlled environments and will not replace human supervision or manual grading in actual classrooms.
- Multi-Subject Expansion – The current project focuses solely on O/L History; other subjects will not be included in this phase.
- Gamification Features – Points, leaderboards, or reward systems will not be implemented.
- Offline Functionality – The system requires internet access for certain features such as cloud-based TTS and dashboard updates; fully offline operation is not supported.
- Live Integration with School Databases – Only locally uploaded Braille sheets and datasets will be used; no direct connection to school grading or attendance databases is included.

3.8 Testing

Testing is a critical phase in the development of EduVisionAI, the automated O/L History correction and feedback system for visually impaired students. Multiple testing approaches will be applied to validate individual modules, integrated functionality, and overall system performance, ensuring reliability, accuracy, and accessibility.

3.8.1 Unit Testing

Purpose:

To validate that each independent module of the system works as intended in isolation before integration.

Scope:

- Braille-to-Text Conversion Module: Verify accurate digitization of uploaded Braille exam sheets into machine-readable text.
- Answer Evaluation Module: Ensure that NLP-based evaluation correctly identifies factual accuracy, missing context, and structural issues in student answers.
- Feedback Generation Module: Test that personalized feedback highlights errors and provides actionable suggestions correctly.
- Text-to-Speech Module: Confirm that audio feedback is correctly generated and understandable for visually impaired learners.

Tools:

- Pytest for automated testing of Python modules.
- Mock Braille Sheets and Sample Answers for controlled test scenarios.

3.8.2 Integration Testing

Purpose:

To verify that all modules work together correctly when combined into the complete system.

Scope:

- Integration of Braille-to-text, answer evaluation, and feedback generation modules to ensure seamless workflow.
- Verification of text and audio feedback delivery through the dashboard.
- Proper storage and retrieval of evaluation results and feedback for progress tracking.
- Real-time communication between backend APIs and frontend dashboard.

Tools:

- Postman for API testing.
- Test dashboards with sample inputs to validate end-to-end functionality.

3.8.3 Performance Testing

Purpose:

To assess the system's responsiveness, accuracy, and computational efficiency.

Metrics:

- Processing Latency: Measure time from Braille upload to generation of textual and audio feedback.
- Evaluation Accuracy: Compare automated scoring and feedback against manually graded samples to ensure alignment with real-world standards.
- System Throughput: Test the number of student responses processed per session under typical hardware configurations.
- Resource Utilization: Monitor CPU, GPU (if used), and memory usage during peak operation.

Tools:

- Python profiling and logging tools for latency and performance measurement.
- Monitoring utilities to track hardware usage during processing.

4 Expected Outcomes

The EduVisionAI system is expected to deliver measurable improvements in exam preparation for visually impaired students by providing accurate automated evaluation, detailed feedback, and accessible multimodal delivery. The outcomes will focus on performance, usability, and educational impact.

Measurable Performance Goals

High Evaluation Accuracy

- Achieve ≥ 90% alignment between automated scoring and expert manual grading of student answers.
- Accurately identify factual errors, missing context, and structural issues in essay-type responses.

Low Processing Latency

- Generate textual and audio feedback within 2–3 seconds per student response.
- Maintain smooth dashboard performance for multiple concurrent student submissions.

Effective Feedback Delivery

- Provide textual feedback that is clear, actionable, and aligned with official marking schemes.
- Deliver audio feedback through TTS that is understandable and accessible for visually impaired students.

Functional Outcomes

- A fully functional Braille-to-text conversion module capable of accurately digitizing Braille exam sheets.
- An automated answer evaluation module using NLP and semantic similarity techniques to assess structured and essay responses.
- A personalized feedback generation module highlighting errors, missing context, and suggestions for improvement.

- A multimodal feedback delivery system providing both text and audio output for inclusive accessibility.

5 Budget and Budget Justification

As the proposed solution, EduVisionAI, is primarily a software-based system, the implementation does not require major hardware purchases. The main expenses are related to cloud services for model training, development, and hosting. Additional costs are expected for project documentation, testing, and dissemination.

Table 2 Budget

Type	**Cost (LKR)**
Internet Use and Web Hosting	10,000.00
Cloud Processing / Training Cost	30,000.00
Publication / Documentation Cost	40,000.00
Stationery	1,000.00
Total	81,000.00

The cost-effective structure, combined with the scalability of the system, makes the project economically viable. The solution can later be commercialized or integrated into existing Intelligent Transportation Systems (ITS) for real-world deployment.

6 Timeline

The project will follow a structured development timeline to ensure systematic progress and timely completion. Initially, the team will focus on system design, including creating a comprehensive architecture diagram, defining module interactions, and selecting the essential technologies and tools for Braille recognition, NLP evaluation, and multimodal feedback delivery.

Next, the implementation phase will begin, emphasizing the development and fine-tuning of the Braille-to-text conversion, automated answer evaluation, and feedback generation modules. Simultaneously, the dashboard interface and text-to-speech integration will be developed to ensure accessible, user-friendly feedback.

The final phase will include extensive testing, performance evaluation, and accessibility validation. Any necessary optimizations will be made, and the team will prepare project documentation for submission and future deployment considerations. Throughout all phases, the project will emphasize careful development, integration, and validation to ensure a reliable and inclusive learning platform.

Figure 3 Timeline

7 Communication Management

The Communications Management Plan is designed to ensure that all team members, as well as the supervisor and co-supervisor, receive the necessary information to effectively perform their roles throughout the project. Successful project execution relies on strategic planning and effective communication. This plan outlines how to communicate with various stakeholders efficiently, detailing the audience, content, format, frequency, and expected outcomes of communications. It also specifies each stakeholder's role, task assignment, and communication strategy based on their influence, interests, and expectations.

Communication Objectives

Proactive communication is essential for project success. Communication should be:

• Adequate: Providing information in the right format and content.

• Specific: Tailored to the targeted audience.

• Sufficient: Including all necessary details.

• Concise: Clear and to the point, avoiding unnecessary repetition.

• Timely: Addressing relevant points at appropriate times.

Communication Media The following media will be utilized for communication throughout the project:

1.Microsoft Teams : For formal updates, documentation, and detailed communications.

2. Documents (MS Word and PowerPoint): For reports, presentations, and detailed project documentation.

3. Phone Calls: For quick discussions and urgent matters.

4. Meetings: Conducted via meeting rooms, conference phones, or Microsoft Teams for in-depth discussions, planning, and progress reviews.

5. Chats (WhatsApp): For informal, quick exchanges and real-time updates.

8 Summary

The proposed EduVisionAI system is an automated, multimodal platform designed to support visually impaired students in preparing for O/L History examinations. By integrating Braille-to-text conversion, NLP-based answer evaluation, and multimodal feedback delivery through both text and audio, the system provides accurate, personalized, and accessible assessment. Real-time dashboard integration allows students and guardians to review results and track progress effectively.

The project follows an Agile development methodology, enabling iterative refinement through unit testing, integration testing, and performance evaluation. Expected outcomes include improved exam readiness, enhanced self-assessment, and equitable access to education. With potential for expansion to other subjects and examination levels, EduVisionAI offers a scalable, cost-effective solution for inclusive learning in Sri Lanka.

9 References

AlSalman, A., Gumaei, A., AlSalman, A., & Al-Hadhrami, S. (2021). A Deep Learning-Based Recognition approach for the conversion of multilingual braille images. *Computers, Materials & Continua/Computers, Materials & Continua (Print)*, *67*(3), 3847–3864. https://doi.org/10.32604/cmc.2021.015614

Nemieboka, T. F., Onyenwe, I. E., & Asogwa, D. C. (2024, January 22). *Development of an NLP-driven computer-based test guide for visually impaired students*. arXiv.org. https://arxiv.org/abs/2401.12375

Gobrecht, A., Tuma, F., Möller, M., Zöller, T., Zakhvatkin, M., Wuttig, A., Sommerfeldt, H., & Schütt, S. (2024, May 7). *Beyond human subjectivity and error: a novel AI grading system*. arXiv.org. https://arxiv.org/abs/2405.04323

Shokat, S., Riaz, R., Rizvi, S. S., Khan, I., & Paul, A. (2022). Characterization of English braille patterns using automated tools and RICA based feature extraction methods. *Sensors*, *22*(5), 1836. https://doi.org/10.3390/s22051836

Braille Works. (2025, April 15). *What is Braille? [Your Guide to Braille] - Braille Works*. https://brailleworks.com/braille-resources/what-is-braille/

Patel, P. K., & Karkare, A. (2019, May 6). *Accessibility evaluation of computer based tests*. arXiv.org. https://arxiv.org/abs/1905.01825